

MARCO CICALÈ BAZTÁN

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Experience

Ph.D. student **Universidad Politécnica de Madrid, IMDEA Software Institute** 2025 - Present

- Full-time enrolled in the *Software, Systems and Computing Ph.D.* program from *Universidad Politécnica de Madrid*.
- Pre-doctoral researcher at the *IMDEA Software Institute* under the supervision of professors JOSÉ F. MORALES and MANUEL HERMENEGILDO; conducting research in the field of *(declarative) programming languages theory, implementation and verification*.
- More specifically, in the analysis, verification and optimization of CHC-based languages (e.g., Prolog) via abstract interpretation.
- Main co-author of [Ciccalè et al., 2025], where we propose a novel approach for the verification of (higher-order) constraint logic programs at compilation time via abstract interpretation, which I presented in *ICLP'25*, and was accepted in the *TPLP* journal.
- Member of the core development team of the Ciao system ([repository](#)).

Research intern **IMDEA Software Institute** 2023 - 2025

- Formalized an abstract interpretation-based theory for the static analysis of higher-order constraint logic programs; and implemented a prototype analysis framework for higher-order constrain logic programs based on program transformation techniques, assertions and goal-dependent abstract interpretation within the Ciao ecosystem for my master's thesis.
- Main author of the Ciao VS Code language extension for my bachelor's thesis.
- It has been adopted as a core tool in the ecosystem of the language, and is broadly used by students at Universidad Politécnica de Madrid in a mandatory course on computational logic.
- Member of the core development team of the Ciao system ([repository](#)).

Projects

BetUnfair 2023

- Engineered and developed a robust, scalable, concurrent and fault-tolerant betting exchange platform.
- Capable of managing 1000 concurrent transactions in 4 seconds while ensuring reliability.
- Took leverage of (pure) functional programming features (like immutability) to ensure a safe concurrent system.
- [Repository](#) | Elixir, OTP Patterns, Functional Programming.

Semantic Web Project 2023

- Transformed and analyzed the energy consumption values of every public building in Madrid.
- Transformed structured data into linked data in RDF format. Created web interface to display the data.
- [Repository](#) | JavaScript, React.js, SPARQL, RDF, OpenRefine.

Education

Universidad {Autónoma,Complutense,Politécnica} de Madrid Madrid, Spain
Masters in Formal Methods in Computer Science and Engineering 2024 - 2025

Universidad Politécnica de Madrid Madrid, Spain
Bachelor in Computer Science 2020 - 2024

Mirabal International School Madrid, Spain
International Baccalaureate | Mathematics and Physics Higher Level 2018 - 2020

Honours

Master's thesis	Universidad Politécnica de Madrid, 2024
Bachelor's thesis	Universidad Politécnica de Madrid, 2024
Formal Methods of Testing	Universidad Complutense de Madrid, 2025
Theory of Programming Languages	Universidad Complutense de Madrid, 2025
AI and Open Science in Research Software Engineering	Universidad Politécnica de Madrid, 2024
Computational Logic	Universidad Politécnica de Madrid, 2023

Publications

[Ciccalè et al., 2025] Ciccalè, M., Jurjo, D., Morales, J. F., López-García, P., and Hermenegildo, M. V. (2025). Hiord#: An Approach to the Specification and Verification of Higher-Order (C)LP Programs. *Theory and Practice of Logic Programming*, 25(4):612–630.